

Core Java – Class Assignment

Assignment Title: Library Management System using Object-Oriented Programming

Topics Covered

- Constructors
- Static Variables
- Static Methods
- Inheritance
- Method Overloading
- Method Overriding
- Arrays of Objects
- Menu-Driven Programming

Problem Statement

Develop a Library Management System in Java that demonstrates constructors, static members, inheritance, method overloading, method overriding, and arrays of objects. The application should be menu-driven and follow object-oriented programming principles.

Part 1: Create the Book Class

- Create a Book class with Book ID, Title, Author, Price, and Quantity.
- Implement a Default Constructor, Parameterized Constructor, and Copy Constructor.
- Implement `displayDetails()`, `updatePrice()`, `purchaseBook()`, and `restockBook()`.
- Create static variables `libraryName` and `totalBooksCreated`.
- Create static methods `displayLibraryInfo()` and `displayTotalBooks()`.

Part 2: Method Overloading

- Overload `updatePrice(double price)` and `updatePrice(double price, double discountPercentage)`.
- Overload `purchaseBook(int quantity)` and `purchaseBook(int quantity, boolean applyDiscount)`.
- Demonstrate all overloaded methods.

Part 3: Inheritance

- Create `PrintedBook` and `EBook` classes that extend `Book`.
- `PrintedBook` should include Number of Pages and Publisher.
- `EBook` should include File Size and File Format.
- Override `displayDetails()` in both subclasses.

Part 4: Method Overriding

- Demonstrate runtime polymorphism using a `Book` reference to access `PrintedBook` and `EBook` objects.

Part 5: Menu-Driven Application

- Store all book objects in an array.
- Implement options to add books, display books, search by ID/title, purchase, restock, update price, display total books created, display library information, and exit.

Submission Guidelines

- Use arrays only. Do not use ArrayList or other Collection Framework classes.
- Follow proper object-oriented programming practices.
- Write clean, properly indented, and well-commented code.
- Submit the complete Java project.